

## 1 Basic Knowledge

**Group**  $(G, *)$ : **G1**: Closure **G2**: Associativity **G3**: Identity element **G4**: Inverse element  $\downarrow$  **Division ring**:  $R^\times = R \setminus \{0\}$ ;  $R^\times$  group (under  $\cdot$ )

**Ring**  $(R, +, \cdot)$ : **R1**:  $(R, +)$  is an abelian group **R2**:  $(R, \cdot)$  is monoid. (no inverse) **R3**: Distributive law (left and right)

**Ideal**  $(I)$ : **I1**:  $\neq \emptyset$  **I2**:  $\forall a, b \in I, a - b \in I$  **I3**:  $\forall r \in R, i \in I, ri, ir \in I$  **Lemma**:  $I = R \Leftrightarrow 1 \in I \Leftrightarrow I \cap R^\times \neq \emptyset$

**Subgroup**  $(H)$ : **H1**:  $e \in H$  **H2**:  $\forall a, b \in H, ab \in H$  **H3**:  $\forall a \in H, a^{-1} \in H$  **Subring**  $(S)$ : **S1**:  $1 \in S$  **S2**:  $\forall a, b \in S, a - b \in S$  **S3**:  $\forall a, b \in S, ab \in S$

**Integral Domain (ID)**: **ID1**: commutative ring **ID2**: no zero divisors **Lemma**: Finite ID  $\Rightarrow$  field ID  $\Rightarrow$  Cancellation Law

**Factor Ring**  $(R/I)$ :  $a + I = b + I \Leftrightarrow a - b \in I$   $(R/I, \oplus, \odot)$  is a ring.  $(1_{R/I} = 1_R + I, 0_{R/I} = 0_R + I, I \text{ ideal})$   $\mathbb{Z}_n \cong \mathbb{Z}/n\mathbb{Z}$

**Ring Homomorphism**:  $\varphi: R \rightarrow S$  s.t. **RH1**:  $\varphi(a + b) = \varphi(a) + \varphi(b)$  **RH2**:  $\varphi(ab) = \varphi(a)\varphi(b)$  **RH3**:  $\varphi(1_R) = 1_S$

**1**:  $\text{Ker}(\varphi)$  is an ideal. **2**:  $\text{Im}(\varphi)$  is a subring. **3**:  $\varphi$  is 1-1  $\Leftrightarrow \text{Ker}(\varphi) = \{0_R\}$ . **FIT**:  $R/\text{Ker}(\varphi) \cong \text{Im}(\varphi)$  (First Isomorphism Theorem)

**Canonical Homomorphism**:  $\text{can}: R \rightarrow R/I$  s.t.  $\text{can}(r) = r + I$  is a onto ring Homomorphism.  $\text{Ker}(\text{can}) = I$   $\updownarrow$  ( $\exists!$  双射)

**Lemma**: If  $\varphi: R \rightarrow S$  ring Homomorphism s.t.  $I \subseteq \text{Ker}(\varphi)$ , then  $\exists! \tilde{\varphi}: R/I \rightarrow S$  s.t.  $\varphi = \tilde{\varphi} \circ \text{can}$  (i.e.  $\tilde{\varphi}(r + I) = \varphi(r)$ )

**Lagrange's Theorem**:  $G$  finite group,  $H \leq G$  subgroup, then  $|H| \mid |G|$

## 2 Types of Ideals (Maximal Ideals $\Rightarrow$ Prime Ideals $\Rightarrow$ Radical Ideals $\Rightarrow$ Ideals)

**Division**: If  $R$  ring, then:  $\forall f, g \in R[x], g$  首相系数可逆  $\Rightarrow \exists q, r \in R[x]$  s.t.  $f(x) = g(x)q(x) + r(x)$  and  $\deg(r) < \deg(g)$  or  $r = 0$ .

**Lemma**: If  $R$  ring,  $g \in R[x], g \neq 0, I = (g)$ .  $\Rightarrow \forall i \in R[x]/I, i = f + I, \deg(f) < \deg(g)$

$g$  首相系数可逆  $\Rightarrow$  If  $a, b \in R[x], \deg(a, b) < \deg(g)$  and  $a + I = b + I$ , then  $a = b$ . (系数也相同)

**Remark**: If  $I = (g), g$  首相系数不可逆: Find a polynomial  $g_1$  s.t.  $g_1 = hg, h \in R[x]$  首相系数可逆, 那么可满足  $^1$  条件,  $^2$  只能 subset 但不能唯一表达.

**Remark**: 有些时候我们能用 gcd, lcm 计算  $\langle a, b \rangle = \langle \gcd(a, b) \rangle$  和  $a \cap b = \langle \text{lcm}(a, b) \rangle$  [If  $R$  commutative ring] 但注意! 要在能够用 Division Algorithm | 有 gcd, lcm 的环里用.

**Principal Ideal**: Ideal  $I$ : iff  $I = aR$  **Prime Ideal**: Ideal  $I \neq R$ :  $ab \in I \Rightarrow a \in I$  or  $b \in I$ . (OR: If  $a \notin I$  and  $b \notin I$ , then  $ab \notin I$ )

**Maximal Ideal**: Ideal  $I \neq R$ : If  $J$  ideal and  $I \subseteq J \subseteq R$ , then  $J = I$  or  $J = R$ .  $\downarrow$   $I$  not radical  $\Leftrightarrow \exists a \notin I$  s.t.  $a^n \in I$  for some  $n \in \mathbb{N}$

**Radical**|**Radical Ideal**:  $I \neq R$ : **Radical**:  $\sqrt{I} = \{r \in R : r^n \in I \text{ for some } n \in \mathbb{N}\}$  **Radical Ideal**: If  $I = \sqrt{I}$   $I \subseteq \sqrt{I}$  always.

**Properties**: Relationship between these kinds of ideals:

- $R$  commutative ring,  $I$  ideal.  $I$  prime  $\Leftrightarrow R/I$  is an ID.  $I$  maximal  $\Leftrightarrow R/I$  is a field
- $R$  ring,  $I_1, I_2, \dots, I_n \triangleleft R, I_1 \cap I_2 \cap \dots \cap I_n = \{0\} \Rightarrow R$  isomorphic to a subring of  $R/I_1 \oplus R/I_2 \oplus \dots \oplus R/I_n$ .  $R \cong R/I_1 \oplus R/I_2 \oplus \dots \oplus R/I_n$  if  $I_i + I_j = R$  for  $i \neq j$
- $R$  ring,  $I$  ideal,  $J$  ideal of  $R/I, T = \{r \in R : r + I \in J\}$  (reps of  $J$ ). Then  $^1 T$ : Ideal of  $R$   $^2 I \subseteq T$   $^3 J = \{t + I : t \in T\}$   $^4 (R/I)/J \cong R/T$   
If  $J = \langle j_1 + I, \dots, j_n + I \rangle$  then  $T = \langle j_1, \dots, j_n \rangle + I = \langle j_1, \dots, j_n, I \rangle$  Moreover, if  $I = \langle i_1, \dots, i_m \rangle$ , then  $T = \langle j_1, \dots, j_n, i_1, \dots, i_m \rangle$
- $R$  commutative ring,  $I$  ideal.  $I$  radical  $\Leftrightarrow R/I$  has no nilpotent elements except 0 (nilpotent:  $r^n = 0$ )
- If  $R$  (commutative with unity), then any ideal  $I \neq R$  is contained in a maximal ideal. If  $r$  is not a unit, then  $1 \notin \langle r \rangle = rR$

## 3 PID | UFD | Euclidean Domain (ED $\Rightarrow$ PID $\Rightarrow$ UFD $\Rightarrow$ ID)

**Irreducible**:  $0 \neq a \in R \setminus R^\times$  is **Irreducible**: If  $a = bc$ , then  $b \in R^\times$  or  $c \in R^\times$ .

**Prime**:  $0 \neq a \in R \setminus R^\times$  is **Prime**: If  $a \mid bc$ , then  $a \mid b$  or  $a \mid c$ . (If  $R$  commutative ring, then prime  $\Rightarrow$  irreducible)

- $F$  field,  $0 \neq g \in F[x]$  non-constant:  $^1 F[x]/\langle g \rangle$  is a field  $\Leftrightarrow ^2 g$  is irreducible in  $F[x] \Leftrightarrow ^3 F[x]/\langle g \rangle$  is a domain
- $F$  field,  $0 \neq g \in F[x]$ :  $g$  is irreducible  $\Leftrightarrow \langle g \rangle$  is prime ideal
- If  $R$  is ID: **Prime  $\Rightarrow$  Irreducible** If  $0 \neq a \in R$ , then:  $a$  prime  $\Leftrightarrow aR$  prime ideal
- If  $R$  is PID,  $a \neq 0$ :  $^1 aR$  is prime ideal  $\Leftrightarrow ^2 a$  is prime  $\Leftrightarrow ^3 aR$  is maximal ideal  $\Rightarrow ^4 a$  is irreducible

**PID**:  $R$  commutative ring is principal ideal domain if:  $\forall I \triangleleft R, I = aR$  for some  $a \in R$ . Theorem:  $\mathbb{Z}, F, F[x]$  is PID.

**ED**:  $R$  is ID.  $R$  is Euclidean Domain if: (Euclidean func)  $\exists v: R \setminus \{0\} \rightarrow \mathbb{N}_0$  s.t.  $\forall a, b \in R, b \neq 0, \exists q, r \in R$  s.t.  $a = bq + r$  and  $r = 0$  or  $v(r) < v(b)$ .

**Common ED**:  $\mathbb{Z}, F[x], \mathbb{Z}[i]$  Gaussian Integers are ED, with  $v(a) = |a|, v(f) = \deg(f), v(a + bi) = a^2 + b^2$  respectively.

**Euclidean Algorithm**: If  $R$  ED,  $a, b \in R \setminus \{0\}$ , then  $\gcd(a, b) = xa + yb$  for some  $x, y \in R$ . (extended of Bezout's Identity)

**UFD (Def 1)**:  $R$  ID is unique factorization domain if: **UFD 1**:  $\forall a \in R \setminus \{0\}, \exists$  irreducibles  $p_1, \dots, p_n \in R$  s.t.  $a = p_1 p_2 \dots p_n$

**UFD 2**: Any other irreducibles  $q_1, \dots, q_m \in R$  s.t.  $a = q_1 q_2 \dots q_m$ , then  $n = m$  and  $p_i = u_i q_i$  for some  $u_i \in R^\times$  after reordering.

**Ascending Chain Condition (ACC)**: A sequence of ideals  $I_1 \subseteq I_2 \subseteq I_3 \subseteq \dots$  satisfies ACC if:  $\exists n \in \mathbb{N}$  s.t.  $I_n = I_{n+1} = I_{n+2} = \dots$

**UFD (Def 2)**:  $R$  ID is UFD if: **UFD 1**: Every sequence of Principal Ideals satisfies Ascending Chain Condition (ACC)

**UFD 2**: If  $a$  irreducible, then  $a$  is prime ( $\Leftrightarrow aR$  is a prime ideal.)

**Causs's Theorem**: If  $R$  is UFD, then  $R[x]$  is UFD.

- Bezout's Identity**:  $R$  is PID, then  $\forall a, b \in R, \exists x, y \in R$  s.t.  $ax + by = \gcd(a, b)$ .
- $R$  UFD:  $a$  prime  $\Leftrightarrow a$  irreducible  $\cdot R$  PID: Every sequence of ideals satisfies ACC.

## 4 Localisation

**Field of Fractions**  $(F_R)$ :  $R$  is ID,  $S = R \setminus \{0\}$ . Def equivalence relation  $\sim$  on  $R \times S$  by:  $(a, b) \sim (c, d) \Leftrightarrow ad = bc$ . (Like  $\frac{a}{b} = \frac{c}{d}$ )

Then the field of fractions  $F_R$  of  $R$ :  $F_R = (R \times S) / \sim = \{\frac{a}{b} : a \in R, b \in S\}$  where  $\frac{a}{b} := \{(a', b') : (a', b') \sim (a, b)\}$

**Operations**:  $\frac{a}{b} + \frac{c}{d} = \frac{ad+bc}{bd}, \frac{a}{b} \cdot \frac{c}{d} = \frac{ac}{bd}$ , Identity:  $0_{F_R} = \frac{0}{1}, 1_{F_R} = \frac{1}{1}$  Inverse:  $-\frac{a}{b} = \frac{-a}{b}, (\frac{a}{b})^{-1} = \frac{b}{a}$  if  $a \neq 0$ .

**Remark**:  $\tau: R \rightarrow F_R$  by  $\tau(r) = \frac{r}{1}$  is a 1-1 ring homomorphism. ps: Isomorphism to  $\{\frac{r}{1} : r \in R\} \subseteq F_R$ .  $F_R$  is a field.

**Multiplicative Set**:  $R$  commutative ring.  $X \subseteq R$  is a multiplicative set if: **1**:  $1_R \in X$  **2**: 乘法封闭 (option \***3**:  $0_R \notin X$ )

**Ring of Fractions**  $(X^{-1}R)$ :  $R$  交换环,  $X \subseteq R$  multiplicative set. Def e.r.  $\sim$  on  $R \times X$  by:  $(a, x) \sim (b, y) \Leftrightarrow \exists s \in X$  s.t.  $s(ay - bx) = 0$ .

Then the ring of fractions  $X^{-1}R$ :  $X^{-1}R = (R \times X) / \sim = \{\frac{a}{x} : a \in R, x \in X\}$  where  $\frac{a}{x} := \{(a', x') : (a', x') \sim (a, x)\}$

**Operations**:  $\frac{a}{x} + \frac{b}{y} = \frac{ay+bx}{xy}, \frac{a}{x} \cdot \frac{b}{y} = \frac{ab}{xy}$ , Identity:  $0_{X^{-1}R} = \frac{0}{1}, 1_{X^{-1}R} = \frac{1}{1}$  Inverse:  $-\frac{a}{x} = \frac{-a}{x}$  If  $a \in X$ , then  $(\frac{a}{x})^{-1} = \frac{x}{a}$ .

**Remark**:  $\varphi: R \rightarrow X^{-1}R$  by  $\tau(r) = \frac{r}{1}$  is a ring homomorphism. ps: Isomorphism to  $\{\frac{r}{1} : r \in R\} \subseteq X^{-1}R$ .  $X^{-1}R$  is a ring.

**Localisation**:  $P$  prime ideal of  $A$ .  $\Rightarrow ^1 S = A \setminus P$  is a multiplicative set.  $^2$  Localisation of  $A$  at  $P$ :  $A_P = S^{-1}A$

**Theorem**:  $A$  commutative ring, then:  $A_P = S^{-1}A$  has only one maximal ideal, which is  $S^{-1}P$ . ps:  $S = A \setminus P$

#### 4.1 Nil and Jacobson Radical Ideals, Local Ring

If  $R$  交换环: **Jacobson Radical**  $J(R) = (\cap \text{maximal ideals})$  [最大 NI]  $\Rightarrow$  **Nil Radical** = **Prime Radical**  $(\cap \text{prime ideals})$  [最大 JRI]

**Nil Ideal:**  $I$  ideal of  $R$  is a *nil ideal* if:  $\forall x \in I, \exists n \in \mathbb{N} \text{ s.t. } x^n = 0$ . i.e. every element of  $I$  is nilpotent. **Prime Radical**  $:= \cap \text{prime ideals}$  ↓

**Nil Radical:**  $I$  ideal of  $R$  is *nil radical* if: It is the largest nil ideal in  $R$ . If  $R$  commutative, then **Nil radical** =  $\cap \text{prime ideals}$

**Jacobson Radical Ideal (JRI):**  $I$  ideal of  $R$  is a *Jacobson radical ideal* if:  $\forall i \in I, 1 - i/1 + i$  is a unit in  $R$ . \*Nil ideal  $\Rightarrow$  JRI(等比)

**Jacobson Radical**  $J(R)$ :  $J(R) := \cap \text{maximal ideals of } R$ .  $\Leftrightarrow J(R)$  is the largest JRI in  $R$ . \*ps: 有如下定义:  $JRI = \sqrt{I} := \cap \{ \text{maximal ideals} \supseteq I \}$

**Local Ring:**  $R$  is a *local ring* if:  $R$  has **unique maximal ideal**  $\Leftrightarrow R/J(R)$  is a field  $A_P$  is a local ring. (A commutative ring,  $P$  prime ideal of  $A$ )

**2nd Def of Radical:**  $R$  commutative ring,  $I$  ideal of  $R$ . Then:  $\sqrt{I} = \cap \{ \text{prime ideals} \supseteq I \}$

#### 5 Nullstellensatz, Variety & Vanishing Ideal

**Weak Nullstellensatz:**  $\mathbb{F}$  field, and  $I \triangleleft \mathbb{F}[x_1, \dots, x_n]$  ideal. Then:  $(\Rightarrow)$  If  $I = \langle x_1 - a_1, \dots, x_n - a_n \rangle, a_i \in \mathbb{F}$ , then  $I$  is maximal ideal.

**Weak Nullstellensatz**  $(\Leftarrow)$  If  $I$  is maximal ideal and  $\mathbb{F}$  is algebraically closed, then  $I = \langle x_1 - a_1, \dots, x_n - a_n \rangle$  for some  $a_i \in \mathbb{F}$ .

**Variety:** Let  $k$  be a field,  $f_1, \dots, f_m \in k[x_1, \dots, x_n]$ . Then: **Variety:**  $V(f_1, \dots, f_m) := \{ (a_1, \dots, a_n) \in k^n : f_i(a_1, \dots, a_n) = 0, \forall i \}$

**Vanishing Ideal:** Let  $k$  be a field,  $X \subseteq k^n$ . Then: **Vanishing Ideal:**  $\mathcal{I}(X) := \{ f \in k[x_1, \dots, x_n] : f(a_1, \dots, a_n) = 0, \forall (a_1, \dots, a_n) \in X \}$

**Nullstellensatz:** Let  $k$  be an algebraically closed field,  $I$  be an ideal of  $k[x_1, \dots, x_n]$ .

1. **The Strong Nullstellensatz:**  $\sqrt{I} = \mathcal{I}(V(I))$  **The Weak Nullstellensatz:**  $V(I) = \emptyset \Leftrightarrow I = k[x_1, \dots, x_n]$  还有一版见上面

2. **Reformation of SN:**  $\cap \text{prime ideals} = \text{Nil radical} = \cap \text{maximal ideals} = \text{Jacobson radical}$

3. **Reformation of WN:** If  $I$  maximal ideal then  $k[x_1, \dots, x_n]/I \cong k$

4. If  $I$  maximal ideal, then  $V(I)$  is a single point. (i.e.  $V(I) = \{ (a_1, \dots, a_n) \}$ )

#### 6 Appendix

**Zorn's Lemma:** If a partially ordered set (i.e.  $\sim \leq$ )  $P$  s.t. <sup>1</sup>  $\forall$  chain in  $P$  has upper bound. <sup>2</sup>  $P \neq \emptyset$ .  $\Rightarrow P$  has at least one maximal element. (i.e.  $\exists m \in P$  s.t.  $\forall p \in P$ , if  $m \leq p$ , then  $m = p$ )